

The 2023 AHRQ National Inpatient Sample (NIS) Quick Start Guide

The NIS Files

The HCUP National Inpatient Sample (NIS) Database is distributed as fixed-width ASCII-formatted data files delivered via secure digital download through [the AHRQ HCUP Central Distributor Online Reporting System \(CDORS\)](#). The files are compressed and encrypted with 7-Zip®. **Users will need the password provided by the HCUP Central Distributor.**

The NIS product is downloaded in a single zipped file for each year which contains several data-related compressed files and accompanying documentation. The NIS is an annual, calendar year file. There are three discharge-level files and one hospital-level file:

NIS Discharge-level files

The unique NIS record number (KEY_NIS) provides the linkage between the discharge-level files.

- **Core File** is a single file containing commonly used data elements (e.g., age, expected primary payer, discharge status, ICD-10-CM/PCS codes, total charges).
- **Severity File** is a single file containing additional data elements to aid in identifying the severity of the condition for a specific discharge.
- **Diagnosis and Procedure Groups File** is a single file containing additional information on the ICD-10-CM diagnoses and ICD-10-PCS procedures that is created by the Agency for Healthcare Research and Quality (AHRQ) software tools.
 - Beginning with data year 2023, data elements derived from the Chronic Condition Indicator Refined (CCIR) for ICD-10-CM are available on the file.

NIS Hospital-level files

The NIS hospital number (HOSP_NIS) provides the linkage between the NIS Core File and the Hospital File. The HOSP_NIS values are reassigned each year, so they cannot be used to link hospitals across years.

- **Hospital File** is a single file containing information on hospital characteristics.

Users interested in applying AHRQ software tools to the NIS for data years including ICD-10-CM/PCS-coded data to produce data elements currently unavailable in the database files may do so by downloading the respective tool(s) from the [Research Tools](#) section of the HCUP User Support (HCUP-US) website. Additionally, users may wish to review the [HCUP Software Tools Tutorial](#), which provides instructions on how to apply the AHRQ software tools to HCUP or other administrative databases.

Computer Specifications Required for Using the NIS

To load and analyze the NIS data on a computer, you will need the following:

- A hard drive with at least 77 gigabytes of space available
- A third-party zip utility such as ZIP Reader, 7-Zip®, SecureZIP®, WinZip®, or Stuffit Expander®
- SAS®, SPSS®, Stata® or similar analysis software.

Decompressing the NIS Files

To extract the data files from the compressed download file, follow these steps:

- 1) Create a directory for the NIS on your hard drive.
- 2) Unzip the compressed NIS product file into the new directory using a third-party zip utility. This will create four compressed, encrypted data-related files in the new directory. You will be prompted to enter the encryption password (sent separately by email) to decrypt the file.

Please note that attempts to unzip encrypted files using the built-in zip utility in Windows® (Windows Explorer) or Macintosh® (Archive Utility) will produce an error message warning of incorrect password and/or file or folder errors. The solution is to use a third-party zip utility.

Third-party zip utilities are available from the following reputable vendors on their official websites.

- ZIP Reader (Windows) (free download offered by the PKWARE corporation)
 - 7-Zip© (Windows) (free download offered by 7-Zip)
 - SecureZIP® for Mac or Windows (free evaluation and licensed/fee software offered by the PKWARE corporation)
 - WinZip (Windows) (evaluation and fee versions offered by the WinZip corporation)
 - Stuffit Expander® (Mac) (free evaluation and licensed/fee software offered by Smith Micro corporation)
- 3) Unzip each of the compressed, encrypted data-related files using the same password and third-party zip utility method. This will place the data-related ASCII files in this same directory by default.

Downloading and Running the Load Programs

Programs to load the data into SAS, SPSS, or Stata, are available on the HCUP User Support website (HCUP-US). The SAS and SPSS programs are available beginning with 1998. The Stata programs begin with 2004. To download and run the load programs, follow these steps:

- 1) Go to the NIS Database Documentation page on HCUP-US at <https://hcup-us.ahrq.gov/db/nation/nis/nisdbdocumentation.jsp>.
- 2) Go to the “File Specifications and Load Programs” section on this page.
- 3) Click on “Nationwide SAS Load Programs”, “Nationwide SPSS Load Programs”, or “Nationwide Stata Load Programs” to go to the corresponding Load Programs page.
- 4) Select the data year and the database (“NIS”) from the drop-down lists on this page. Or you may select “NIS Load All Years” to obtain a zipped file with all load programs for multiple years at once.
- 5) Select and save the load programs you need. The **load programs are specific to the data year**. For example, the load program for the 2023 NIS Core File is found under the link “SAS NIS 2023 Core File” in the list generated by selecting “2023” and “NIS.” Save the load programs into the same directory as the NIS ASCII files on your computer.
- 6) Edit and run the load programs as appropriate for your computing environment to create the analysis files. For example, modify the directory paths to point to the location of your input and output files.

NIS Documentation

Comprehensive documentation for the NIS files is available on the HCUP-US website (<https://hcup-us.ahrq.gov/db/nation/nis/nisdbdocumentation.jsp>). Refer to these resources to understand the structure and content of the NIS and to aid in using the database. For example, the *Introduction to the NIS* includes information on how to use the NIS data and discharge weights for analysis. The [HCUP Methods Series Report Using the HCUP National Inpatient Sample to Estimate Trends](#) includes details regarding revised weights that account for design changes and other recommendations for trends analysis.

For information on modifications to the NIS in data year 2023 due to changes in the participating states relative to prior years of the NIS, see the *Introduction to the NIS, 2023* ([PDF](#)).

HCUP Online Tutorials

For additional assistance, AHRQ has created the HCUP Online Tutorial Series, a series of free, interactive courses that provide information on using HCUP data and tools and training on technical methods for conducting research with HCUP data. Topics include an [HCUP Overview Course](#) and these tutorials:

- The [Load and Check HCUP Data](#) tutorial provides instructions on how to unzip (decompress) HCUP data, save it on your computer, and load the data into a standard statistical software package. This tutorial also describes how to verify that the data have loaded correctly.
- The [HCUP Sampling Design](#) tutorial is designed to help users learn how to account for sample design in their work with HCUP nationwide databases.
- The [Producing National HCUP Estimates](#) tutorial is designed to help users understand how three of the nationwide databases – the NIS, NEDS, and KID – can be used to produce national and regional estimates.
- The [Calculating Standard Errors](#) tutorial shows how to accurately determine the precision of the estimates produced from the HCUP nationwide databases. Users will learn two methods for calculating standard errors for estimates produced from the HCUP nationwide databases.
- The [Multi-year Analysis](#) tutorial presents solutions that may be necessary when conducting analyses that span multiple years of HCUP data.
- The [HCUP Software Tools Tutorial](#) provides instructions on how to apply the AHRQ software tools to HCUP or other administrative databases.

New tutorials are added periodically. The Online Tutorial Series is located on the HCUP-US website at https://hcup-us.ahrq.gov/tech_assist/tutorials.jsp.

HCUP Central Distributor

If you need assistance for issues related to unzipping the files or have questions about passwords, please contact the HCUP Central Distributor and User Support team.

- E-mail: HCUP-RequestData@ahrq.gov
- You may also refer to the [HCUP Load and Check Tutorial](#) for troubleshooting tips.

HCUP User Support

For questions about using the HCUP databases (including loading files), software tools, supplemental files, and other HCUP products, or about data use restrictions and publishing with the data, please review the [HCUP Frequently Asked Questions](#) or contact HCUP User Support:

- E-mail: hcup@ahrq.gov
- Requirements for Publishing with HCUP Data: <https://hcup-us.ahrq.gov/db/publishing.jsp>

For general information about HCUP data, tools and products, please visit the HCUP-US website at <https://hcup-us.ahrq.gov/>